Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Cost-Efficient and Continually Learning-Based Field-Adaptive AI for Acoustic Anomaly Detection			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Measuring Instruments	Safety Monitoring/Diagnosis Control Instruments
Concept of Proposed Technology	Current acoustic anomaly detection systems often require repeated data collection and re-training for each new site, and their performance degrades over time as acoustic conditions change, resulting in high deployment and maintenance costs. Such sensitivity to environmental variation and data scarcity has been consistently reported in recent studies on few-shot and one-shot acoustic anomaly detection, as well as in surveys of industrial audio-based fault diagnosis systems [1–4]. To address these limitations, this technology proposes a cost-efficient and continually learning field-adaptive AI for acoustic anomaly detection, which can rapidly adapt to new environments and maintain stable performance over time. By combining fast adaptation and continual learning within a unified operational pipeline, the proposed approach minimizes repeated retraining and manual intervention, enabling scalable and sustainable deployment.			

Rationale for Joint R&D Project	Field-adaptive and continually learning AI systems must be validated under diverse environmental and operational conditions to ensure true generalization. This cannot be achieved through single-country or single-site experiments alone. Recent studies on fast adaptation and continual learning have primarily demonstrated their effectiveness in controlled or limited experimental settings, highlighting the lack of systematic validation across heterogeneous real-world environments [1,6]. The international collaboration allows systematic evaluation across heterogeneous environments, enabling the development of more robust, transferable, and cost-efficient adaptation mechanisms. This joint approach significantly reduces technical uncertainty and enhances the global applicability of the proposed technology.
Key Words	1. Field-Adaptive Learning 2. Continual Learning 3. Cost-Efficient Adaptation 4. Acoustic Anomaly Detection

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 3	End of the Project	Level 7
R&D Objectives	The objective of this project is to develop a cost-efficient and continually learning field-adaptive AI for acoustic anomaly detection that can be rapidly deployed and sustainably operated across diverse environments. Specifically, the project aims to *Objective 1.* Design **fast and low-cost adaptation algorithms** that enable stable performance with minimal labeled data *Objective 2.* Develop **continual learning mechanisms** that maintain long-term reliability under evolving acoustic conditions without catastrophic forgetting [5,6] *Objective 3.* Integrate these core technologies into an **operational learning pipeline** that minimizes manual intervention, repeated retraining, and deployment overhead			
R&D Target Specifications	The target performance specifications of this project are as follows: 1. Reduce the amount of labeled data required for environment adaptation by at least 70% compared to conventional fine-tuning approaches. 2. Improve anomaly detection performance in unseen environments by achieving a minimum 15% increase in F1-score relative to non-adaptive baseline models. 3. Reduce false alarm rates by at least 40%, enhancing practical usability and operational reliability. 4. Enable stable long-term operation through continual learning, maintaining performance degradation within 5% over extended deployment periods without manual retraining. 5. Shorten adaptation and stabilization time after deployment to within 24 hours using automated adaptation mechanisms.			
R&D Plan	**Phase 1: Core Algorithms for Cost-Efficient and Fast Adaptation (TRL 3 → 4)** In Phase 1, the project focuses on developing core algorithms for cost-efficient and fast adaptation of acoustic anomaly detection models in new environments. The main research question is to identify which components of the model must be adapted and which can remain invariant across domains. To address this, the project will explore few-shot learning, weakly supervised adaptation, and parameter-efficient tuning strategies, such as representation alignment and lightweight fine-tuning mechanisms. Confidence-aware updating and partial model adaptation will be investigated to minimize the amount of labeled data and computational cost required for stabilization. The main milestone of this phase is the demonstration of stable and reproducible adaptation performance under simulated domain shifts and public benchmark datasets, validating that the proposed methods can achieve rapid stabilization with significantly reduced data requirements. The preliminary investigation will therefore draw on insights from prior few-shot and one-shot acoustic anomaly detection studies that emphasize data efficiency and rapid adaptation [2,3]. **Phase 2: Continual Learning for Long-Term Stability (TRL 4 → 5)** While Phase 1 focuses on rapid and cost-efficient adaptation to new environments, Phase 2 extends this capability to long-term operation by addressing stability and knowledge retention under continuously evolving acoustic conditions. The project will investigate methods for mitigating catastrophic forgetting, including regularization-based memory preservation, replay-free incremental learning, and lightweight memory			

	mechanisms. In parallel, drift-aware updating strategies will be developed using unsupervised change-point detection, distribution shift detection, and reliability-based monitoring. Furthermore, confidence-aware self-updating and weakly supervised self-training will be explored to enable the system to continuously improve without requiring explicit manual labeling. The milestone of this phase is the implementation of a continual learning framework that preserves past knowledge while incrementally adapting to new acoustic conditions, maintaining stable detection performance over extended time periods. **Phase 3: Operational Learning Pipeline and Field Validation (TRL 5 → 7)** Building upon the adaptation and continual learning mechanisms developed in Phase 1 and Phase 2, Phase 3 aims to integrate these core technologies into a unified operational pipeline and validate them in real-world deployment scenarios. In this phase, the developed adaptation and continual learning algorithms will be implemented as a unified operational learning pipeline that automates the processes of data ingestion, reliability estimation, drift detection, adaptation triggering, and performance monitoring. The system will support low-touch deployment by combining fast adaptation modules from Phase 1 and continual learning mechanisms from Phase 2 into a closed-loop learning architecture. Field-level validation will be conducted in real-world environments provided by the Korean SME partner. The system will be evaluated using practical KPIs, such as adaptation time, reduction of manual intervention, false alarm rate, and long-term stability under real operational conditions. The final milestone is the demonstration of a scalable, cost-efficient, and continually learning acoustic anomaly detection system that can be sustainably deployed across diverse environments.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Fraunhofer IDMT) will lead the core algorithmic research and system-level design of the proposed technology. Its primary role is to develop and validate the fundamental learning mechanisms that enable cost-efficient adaptation and continual learning in acoustic anomaly detection. Specifically, the institute will: 1. Design and implement core algorithms for fast and low-cost adaptation, including few-shot learning, parameter-efficient tuning, and representation alignment. 2. Develop continual learning mechanisms that prevent catastrophic forgetting and ensure long-term stability under evolving acoustic conditions. 3. Formulate drift detection and reliability-aware update strategies to enable autonomous and robust model maintenance. 4. Establish a unified operational learning pipeline architecture, integrating adaptation, monitoring, and self-updating modules. 5. Conduct systematic benchmarking, ablation studies, and long-term evaluations to ensure scientific validity and generalizability. In addition, Fraunhofer IDMT will provide research infrastructure, computing resources, and expertise in acoustic signal processing and machine learning, and will be responsible for disseminating scientific outcomes through publications, patents, and technical documentation.
Expected R&R of Korean SME Partner	The Korean SME partner will play a key role in real-world deployment, field validation, and commercialization-oriented development of the proposed technology. While the research institute focuses on core algorithmic innovation, the SME will ensure that the developed methods are practical, scalable, and aligned with market requirements. Specifically, the SME partner will: 1. Provide real-world acoustic data and diverse deployment environments to support realistic evaluation of adaptation and continual learning methods. 2. Lead the field-level implementation and testing of the proposed system, including installation, operation, and maintenance. 3. Define practical performance requirements and operational KPIs, such as false alarm tolerance, adaptation time, and maintenance cost. 4. Support the development of user-oriented interfaces and deployment workflows to minimize manual intervention. 5. Contribute to productization and commercialization planning, including integration into existing solutions and market validation. Through these roles, the SME partner will ensure that the proposed technology is not only scientifically sound but also economically viable and deployable at scale.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver both scientific advancements and practical industrial value. For AI researchers and system developers, the project will establish novel frameworks for cost-efficient adaptation and continual learning in acoustic anomaly detection, contributing to the advancement of adaptive and lifelong AI systems. For SMEs and solution providers, the developed technology will significantly reduce deployment and maintenance costs, enabling faster and more scalable adoption of acoustic AI solutions. The unified operational learning pipeline will allow system integrators to deploy and maintain AI systems across multiple sites with minimal manual intervention, shortening time-to-market and lowering operational risks. For end users and industrial operators, the proposed system will provide more reliable and sustainable AI-based monitoring, leading to improved operational safety, reduced downtime, and enhanced productivity. At the broader industrial and economic level, this international collaboration will strengthen global competitiveness, promote technology transfer, and contribute to the growth of AI-driven services and data-centric industries.

6. References

(1) Chen, Bingqing, Luca Bondi, and Samarjit Das. "Learning to adapt to domain shifts with few-shot samples in anomalous sound detection." 2022 26th International Conference on Pattern Recognition (ICPR). IEEE, 2022.

(2) Tanaka, Ryosuke, and Satoshi Tamura. "Few-shot anomalous sound detection based on anomaly map estimation using pseudo abnormal data." ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2024.

(3) Koizumi, Yuma, et al. "Spidernet: Attention network for one-shot anomaly detection in sounds." ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2020.

(4) Tang, Lili, et al. "A survey of mechanical fault diagnosis based on audio signal analysis." Measurement 220 (2023): 113294.

(5) Kirkpatrick, James, et al. "Overcoming catastrophic forgetting in neural networks." Proceedings of the national academy of sciences 114.13 (2017): 3521-3526.

(6) De Lange, Matthias, et al. "A continual learning survey: Defying forgetting in classification tasks." IEEE transactions on pattern analysis and machine intelligence 44.7 (2021): 3366-3385.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization